

MH2100 Calculus III

Midterm I (Solutions)

AY 2022/2023, Semester 1

Quantitative Research Society @NTU

September 2022

Overview

- **Coverage:** parametric curves and tangent lines; limits and continuity; partial and directional derivatives; tangent planes, the Chain Rule, implicit differentiation, and critical points.
- **Format:** three questions, 50 marks in total, 90 minutes.
- **Current administration (AY2026/2027):** Tuesday 6 October 2026, 16:30–18:20, venue TBA. One double-sided typed A4 sheet is permitted; no attachments or sticky notes are permitted.
- **Source:** MH2100 Midterm I, AY2022/2023 Semester 1. Every question below is transcribed from the official paper. Solutions and marking schemes are unofficial QRS@NTU material.

Instructions to candidates (verbatim)

1. DO NOT TURN OVER PAPER UNTIL INSTRUCTED.
2. This midterm paper contains THREE (3) questions.
3. Answer ALL questions. The marks for each question are indicated at the beginning of each question.
4. Candidates can write anywhere on this midterm paper.
5. ONE two-sided A4 piece of paper with handwritten notes is permitted.

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Question 1**(17 Marks)****QUESTION 1.**

- (i) [3 marks] Find parametric equations for the line that is tangent to the curve parametrised by

$$\mathbf{r}(t) = \begin{pmatrix} t^2 - 2t + 1 \\ \sin(t^2 - 1) \end{pmatrix},$$

at the point $(0, 0)$.

- (ii) [3 marks] At the points defined by $t = a$ the tangent line to the curve parametrised by

$$\begin{cases} x(t) = 2t, \\ y(t) = 3t^2 + t, \\ z(t) = -3t^2 - t \end{cases}$$

contains the point $(0, -12, 12)$. Find all such values of a .

- (iii) [3 marks] Evaluate the following limit or show that the limit does not exist.

$$\lim_{(x,y) \rightarrow (0,0)} \frac{3xy - y}{x^2 - 3x + y^2 + 7y + 1}.$$

- (iv) [4 marks] At the point defined by $t = a$ the tangent vector of the curve parametrised by

$$\mathbf{r}(t) = \begin{pmatrix} \cos(3t^2) \\ t \\ t^2 + 2t \end{pmatrix},$$

is a scalar multiple of the gradient vector $\nabla f(1, 2, 3)$ where $f(x, y, z) = x^2 + y^2 - 2x - 2yz$. Find the value of a .

- (v) [4 marks] Find the directional derivative of the function

$$f(x, y) = \begin{cases} \frac{x^3 y^2 - 3y^5}{x^4 + 2x^2 y^2 + y^4}; & \text{if } x^2 + y^2 \neq 0, \\ 0; & \text{if } x^2 + y^2 = 0, \end{cases}$$

in the direction of the vector $\langle 1, -1 \rangle$ at the point $(0, 0)$.

Solution**Method 1: Direct computation**

- (i) The point condition gives $(t - 1)^2 = 0$, hence $t = 1$; independently, $\sin(1^2 - 1) = 0$. Differentiate:

$$\mathbf{r}'(t) = \langle 2t - 2, 2t \cos(t^2 - 1) \rangle, \quad \mathbf{r}'(1) = \langle 0, 2 \rangle.$$

Thus the tangent line is

$$\boxed{\mathbf{l}(s) = \langle 0, 0 \rangle + s \langle 0, 2 \rangle, \quad s \in \mathbb{R},}$$

equivalently $\boxed{x = 0, \quad y = 2s.}$

(ii) At $t = a$,

$$\mathbf{r}(a) = \langle 2a, 3a^2 + a, -3a^2 - a \rangle, \quad \mathbf{r}'(a) = \langle 2, 6a + 1, -6a - 1 \rangle.$$

If $(0, -12, 12) = \mathbf{r}(a) + s\mathbf{r}'(a)$, the first component gives $s = -a$. Expanding the second gives

$$-12 = 3a^2 + a - a(6a + 1) = -3a^2,$$

so $a = \pm 2$. The third component independently checks as $3a^2 = 12$.

$$\boxed{a = -2 \text{ or } a = 2.}$$

(iii) The denominator tends to $1 \neq 0$, so the quotient is continuous at the point. Direct substitution yields

$$\boxed{0}.$$

(iv)

$$\mathbf{r}'(t) = \langle -6t \sin(3t^2), 1, 2t + 2 \rangle, \quad \nabla f(x, y, z) = \langle 2x - 2, 2y - 2z, -2y \rangle.$$

Hence $\nabla f(1, 2, 3) = \langle 0, -2, -4 \rangle$. From $\mathbf{r}'(a) = \lambda \nabla f(1, 2, 3)$, the second component gives $\lambda = -1/2$; the third then gives $2a + 2 = 2$. Thus $a = 0$, and the first component also vanishes. Therefore $\boxed{a = 0.}$

(v) Let $\mathbf{u} = \langle 1, -1 \rangle / \sqrt{2}$. On $(x, y) = t\mathbf{u}$, expand independently:

$$x^3 y^2 - 3y^5 = \frac{t^5}{4\sqrt{2}} + \frac{3t^5}{4\sqrt{2}} = \frac{t^5}{\sqrt{2}},$$

$$x^4 + 2x^2 y^2 + y^4 = \frac{t^4}{4} + \frac{t^4}{2} + \frac{t^4}{4} = t^4.$$

So $f(t\mathbf{u}) = t/\sqrt{2}$ and $f(0, 0) = 0$. Therefore

$$D_{\mathbf{u}}f(0, 0) = \lim_{t \rightarrow 0} \frac{f(t\mathbf{u}) - f(0, 0)}{t} = \boxed{\frac{1}{\sqrt{2}}}.$$

UNOFFICIAL QRS marking scheme (17 marks)

- (i) **(3 marks)** 1: identifies $t = 1$; 1: finds the tangent vector; 1: gives the line.
- (ii) **(3 marks)** 1: writes the tangent line; 1: solves $a^2 = 4$; 1: gives both values and checks the remaining component.
- (iii) **(3 marks)** 1: identifies a nonzero denominator; 1: justifies substitution; 1: obtains 0.
- (iv) **(4 marks)** 1: differentiates $\mathbf{r}$; 1: computes the gradient; 1: imposes proportionality; 1: verifies $a = 0$.
- (v) **(4 marks)** 1: normalizes the direction; 1: substitutes the path; 1: expands both expressions; 1: evaluates the limit.

Question 2**(16 Marks)****QUESTION 2.**

- (i) [4 marks] Evaluate each of the following limits or show that the limit does not exist. Justify your answers.

(a)

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^3 y}{x^3 y + (x - y)^4}$$

(b)

$$\lim_{(x,y) \rightarrow (1,0)} \frac{7x^2 - 7}{x^2 + y^2 - 1}$$

- (ii) [4 marks]

(a) Find k such that the function

$$f(x, y) = \begin{cases} \frac{x^4 + 3x^2 - xy^3 + 3y^2}{x^2 + y^2} & \text{if } x^2 + y^2 > 0; \\ k & \text{if } x^2 + y^2 = 0 \end{cases}$$

is continuous. (No justification is needed).

(b) For the k found in Part (a), prove that

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^4 + 3x^2 - xy^3 + 3y^2}{x^2 + y^2} = k.$$

- (iii) [8 marks] Define

$$f(x, y) = \begin{cases} \frac{x^3 - xy^2}{x^2 + y^2}; & \text{if } x^2 + y^2 \neq 0, \\ 0; & \text{if } x^2 + y^2 = 0. \end{cases}$$

- (a) Is f continuous at $(0, 0)$? Justify your answer.
 (b) Find $f_x(0, 0)$ and $f_y(0, 0)$.
 (c) Is f differentiable at $(0, 0)$? Justify your answer.

Solution**Method 1: Direct path tests, bounds, and definitions**

- (i) (a) Along $y = x$ the quotient is 1, while along $y = 0$ it is 0. Therefore

the limit does not exist.

- (b) Along $x = 1$ the quotient is 0. Along $y = 0$, factoring gives $7(x - 1)(x + 1)/((x - 1)(x + 1)) = 7$. Thus the limit does not exist.

- (ii) (a) The required continuous value is $k = 3$.

(b) Put $r = \sqrt{x^2 + y^2}$. Expand

$$\frac{x^4 + 3x^2 - xy^3 + 3y^2}{x^2 + y^2} = 3 + \frac{x^4 - xy^3}{x^2 + y^2}.$$

The remainder satisfies

$$\left| \frac{x^4 - xy^3}{x^2 + y^2} \right| \leq \frac{|x|^4 + |x||y|^3}{r^2} \leq \frac{2r^4}{r^2} = 2r^2 \rightarrow 0.$$

Hence the requested limit is $\boxed{3 = k}$.

(iii) (a)

$$|f(x, y)| = \left| x \frac{x^2 - y^2}{x^2 + y^2} \right| \leq |x| \rightarrow 0.$$

Since $f(0, 0) = 0$, $\boxed{f \text{ is continuous at } (0, 0)}$.

(b) From $f(h, 0) = h$ and $f(0, h) = 0$ for $h \neq 0$,

$$\boxed{f_x(0, 0) = 1, \quad f_y(0, 0) = 0.}$$

(c) The only possible derivative is $L(x, y) = x$. On $(x, y) = (t, t)$, $f(t, t) = 0$, so

$$\frac{f(t, t) - f(0, 0) - L(t, t)}{\sqrt{t^2 + t^2}} = \frac{-t}{\sqrt{2}|t|},$$

which does not tend to 0. Thus $\boxed{f \text{ is not differentiable at } (0, 0)}$.

UNOFFICIAL QRS marking scheme (16 marks)

(i) (4 marks)

(a) (2 marks) 1: $y = x$; 1: $y = 0$ and conclusion.

(b) (2 marks) 1: two valid paths; 1: nonexistence conclusion.

(ii) (4 marks)

(a) (1 mark) $k = 3$.

(b) (3 marks) 1: separates 3; 1: bounds the remainder; 1: concludes the limit.

(iii) (8 marks)

(a) (2 marks) 1: valid bound; 1: continuity.

(b) (3 marks) 1: x -axis quotient; 1: y -axis quotient; 1: stated partials.

(c) (3 marks) 1: derivative L ; 1: path calculation; 1: failure of the remainder condition.

Question 3**(17 Marks)****QUESTION 3.**

- (i) [3 marks] Find an equation for the tangent plane to the surface given by the equation $z = \ln(8x^2 + 9y^2 + 1)$ at the point $(0, 0, 0)$.

- (ii) [3 marks] Use the Chain Rule to find $\frac{\partial u}{\partial y}$ at $(x, y, z) = (3, 2, 0)$ for the function

$$u(p, q, r) = e^{pq} \cos(r),$$

where

$$p = \frac{1}{x}, \quad q = x^2 \ln y, \quad r = z.$$

- (iii) [3 marks] Assume that z is uniquely determined by the equation

$$9 \sin(x + y) + 4 \sin(x + z) + 3 \sin(y + z) = 0.$$

Find the value of $\frac{\partial z}{\partial x}$ at the point $(2\pi, -\pi, 4\pi)$.

- (iv) [3 marks] In which directions is the directional derivative of

$$f(x, y) = \sin(xy) - x^2 + xy$$

at $(\pi, 0)$ equal to zero? Write the unit vector for each direction.

- (v) [5 marks] Let

$$f(x, y) = 2y^3 + x^2 - 6xy - 8x + 4.$$

- (a) Find all critical points of $f(x, y)$.
 (b) For each critical point found in Part (a), determine whether or not it is a local max/min or a saddle point.

Solution**Method 1: Direct differentiation and classification**

- (i) Let $g(x, y) = \ln(8x^2 + 9y^2 + 1)$. Then

$$g_x = \frac{16x}{8x^2 + 9y^2 + 1}, \quad g_y = \frac{18y}{8x^2 + 9y^2 + 1}.$$

At the stated point, both slopes and $g(0, 0)$ are 0. Hence the tangent plane is $\boxed{z = 0}$.

- (ii) The Chain Rule is $u_y = u_p p_y + u_q q_y + u_r r_y$. Here $p_y = r_y = 0$, $q_y = x^2/y$, and $u_q = pe^{pq} \cos r$, so

$$u_y = pe^{pq} \cos r \frac{x^2}{y}.$$

At $(3, 2, 0)$, $p = 1/3$, $q = 9 \ln 2$, $r = 0$, and $pq = 3 \ln 2$. Hence

$$\boxed{u_y = \frac{1}{3} e^{3 \ln 2} \cdot \frac{9}{2} = \frac{1}{3} \cdot 8 \cdot \frac{9}{2} = 12.}$$

(iii) Let $F(x, y, z) = 9 \sin(x + y) + 4 \sin(x + z) + 3 \sin(y + z)$. Then $F_x + F_z z_x = 0$, with

$$F_x = 9 \cos(x + y) + 4 \cos(x + z), \quad F_z = 4 \cos(x + z) + 3 \cos(y + z).$$

At the point, the three cosines are $-1, 1, -1$, respectively. Thus $F_x = -5$, $F_z = 1$, and $z_x = -F_x/F_z = 5$.

(iv)

$$f_x = y \cos(xy) - 2x + y, \quad f_y = x \cos(xy) + x.$$

Thus $\nabla f(\pi, 0) = 2\pi \langle -1, 1 \rangle$. A unit vector $\langle u_1, u_2 \rangle$ gives zero derivative exactly if $u_1 = u_2$ and $u_1^2 + u_2^2 = 1$. Therefore

$$\mathbf{u} = \frac{1}{\sqrt{2}} \langle 1, 1 \rangle \quad \text{or} \quad \frac{1}{\sqrt{2}} \langle -1, -1 \rangle.$$

(v) (a)

$$f_x = 2x - 6y - 8, \quad f_y = 6y^2 - 6x.$$

The equations give $x = 3y + 4 = y^2$, hence $(y - 4)(y + 1) = 0$. The critical points are $(16, 4)$ and $(1, -1)$.

(b)

$$f_{xx} = 2, \quad f_{xy} = -6, \quad f_{yy} = 12y, \quad D = f_{xx}f_{yy} - f_{xy}^2 = 24y - 36.$$

At $(16, 4)$, $D = 60 > 0$ and $f_{xx} > 0$, so it is a local minimum. At $(1, -1)$, $D = -60 < 0$, so it is a saddle point. Thus

$$(16, 4) \text{ is a local minimum; } (1, -1) \text{ is a saddle point.}$$

UNOFFICIAL QRS marking scheme (17 marks)

- (i) **(3 marks)** 1: both partials; 1: evaluates slopes; 1: plane $z = 0$.
- (ii) **(3 marks)** 1: Chain Rule; 1: intermediate derivatives; 1: obtains 12.
- (iii) **(3 marks)** 1: F_x, F_z ; 1: evaluates cosines; 1: obtains 5.
- (iv) **(3 marks)** 1: gradient; 1: orthogonality condition; 1: both unit vectors.
- (v) **(5 marks)**
 - (a) **(2 marks)** 1: solves first-derivative equations; 1: both points.
 - (b) **(3 marks)** 1: Hessian and discriminant; 1: minimum classification; 1: saddle classification.